

First steps towards a facial recognition login system for a professional networking company

Sofía Jimeno Lucía
Master in Computer Vision - IMCV
Universidade do Porto
up202200682@edu.fe.up.pt

Nils Meins
Supervisor of the project at XING company
Hamburg, Germany
nils.meins@xing.com

DISCLAIMER

The views, findings and conclusions expressed in this academic report are solely those of the authors and do not represent the official position of XING or any company associated with New Work SE. The report is for academic purposes only, documenting the authors' experience, and should not be considered an endorsement or professional advice. It does not contain any confidential information and does not include any user data.

I. INTRODUCTION

In today's digital era, with the increasing reliance on biometric authentication methods, the demand for secure, efficient, and reliable face recognition systems is higher than ever. This is particularly pertinent for applications like facial login systems on career-oriented social networking sites. These platforms, which store sensitive personal and professional information, require robust security mechanisms to protect user data from unauthorized access.

The goal of this project was to explore State-of-the-Art (SoTA) methods to develop a facial login system for a career-oriented social networking site (XING), by developing a model as outlined in the Face Recognition Challenge in the Era of Synthetic Data (FRCSyn) using synthetic data.

XING [2] is a career-oriented social networking platform designed to help professionals connect, network, and advance their careers. It allows users to create detailed profiles, join industry-specific groups, and discover job opportunities tailored to their skills and interests. XING is particularly popular in German-speaking countries, providing tools for professional development, business networking, and knowledge sharing. It also features events and webinars to facilitate continuous learning and career growth.

The CVPR 2024 Challenge [1] "...intends to explore the application of synthetic data to the field of face recognition in order to find solutions to the current limitations in the technology, for example, in terms of privacy concerns associated with real data, bias in demographic groups (e.g., ethnicity and gender), and lack of performance in challenging conditions such as large age gaps between enrolment and testing, pose variations, occlusions, etc."

II. DATASETS

As explained in the basics of the challenge, the training part of the model should be trained using synthetic data, for which the DCFace dataset [8] was selected, available on GitHub and specifically designed for face recognition research using deep convolutional neural networks (DCNNs). It consists of a diverse collection of face images annotated with identity labels, capturing variations in pose, lighting and expression to ensure robust model training. The images are pre-processed for alignment and cropping to provide a standardised input to the neural networks. The directory structure, with separate folders for each individual's images, consists of over 10.000 folders or subjects, each containing 55 images. Some examples are shown in Figure 1. It is important to point out the difficulty of this dataset due to the generated images, they may have some distortions in the factions of the person and also the fact that the images from the same folder should be similar to each other, but this is usually not fulfilled.

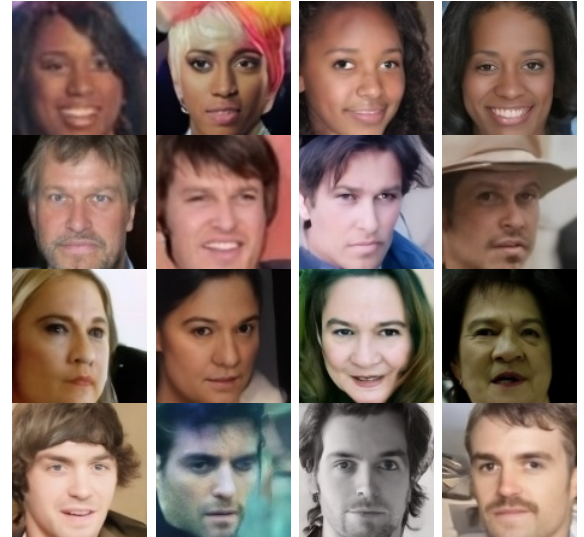


Fig. 1. Examples DCFace Dataset

On the other hand, for the testing part, a real face dataset was used, specifically the CASIA-WebFace [8] dataset, created by the Chinese Academy of Sciences' Institute of Automation. It consists of over 500.000 images of

approximately 10.575 subjects, with diverse representations of demographics, ethnicity and age. The dataset includes annotations such as facial landmarks, head poses and other attributes, making it valuable for tasks such as face recognition, verification and attribute analysis.



Fig. 2. Examples CASIA-WebFace Dataset

Finally, Fig. 3 is a profile picture of Sofía, one of the authors of this report, and was used to test all the experiments during the external practices. To comply with the General Data Protection Regulation (GDPR) and prevent any potential issues, it was decided to only use our own pictures or publicly available pictures. Therefore, it is important to emphasize again that no user data was utilized in any of the tests performed.



Fig. 3. Profile picture of one of the authors

III. TECHNICAL APPROACHES

A. Triplet Loss

Triplet loss is a deep learning loss function that is particularly useful in training neural networks for tasks that require learning embeddings, such as face recognition or image similarity. The goal is to ensure that the distance between an Anchor and a Positive sample (a sample of the same class) is less than the distance between the Anchor and a Negative

sample (a sample of a different class) by a margin. The process used to implement Triplet Loss is the following:

1) *Batch Processing*: Generate batches of triplets from the training dataset: Anchor, the reference sample; Positive, a sample from the same class as the anchor. Negative, a sample from a different class than the Anchor. The total number of subjects used was 5.000 where each batch had a size of 128 triplets to be processed in parallel.

2) *Embedding Vectors*: Feeding each image in the triplet (Anchor, Positive, Negative) through the Siamese Network, which has shared weights, and obtaining embeddings for each image of the triplet. These embeddings capture the essential features of the inputs.

3) *Loss Calculation*: Calculate the Euclidean distances between embeddings: distance between Anchor and Positive, and distance between Anchor and Negative. Compute the loss for each triplet using the formula of Figure 4:

$$\mathcal{L} = \max(d(a, p) - d(a, n) + \text{margin}, 0)$$

Fig. 4. Triplet Loss Function

where:

- $d(x, y)$ is the distance metric (as mention before, Euclidean distance in our case) between samples x and y .
- α is a margin that is enforced between positive and negative pairs. It is a positive scalar that helps in ensuring that the distance between the anchor and the negative sample is greater than the distance between the anchor and the positive sample by at least α .

4) *Backpropagation and Parameters Updates*: Compute gradients of the triplet loss with respect to the network parameters and use an optimization algorithm, in our case was Adam (0.0001), to update the network's weights based on the computed gradients. Repeat the process for each batch, iteratively optimizing the network until convergence.

By focusing on these core steps, the Siamese Network is trained to produce embeddings where similar images are close in the embedding space, and dissimilar images are far apart.

B. Semi-hard Triplet Loss

Semi-hard triplet loss is a variation of the triplet loss that focuses on selecting more challenging triplets for training. The key idea is to choose negative samples that are neither too easy nor too hard. Specifically, semi-hard negatives are those that are closer to the Anchor than the Positive sample, but still violate the margin condition to some extent. For this case, *TripletSemiHardLoss* is used as implementation of the triplet loss function provided by Tensorflow's high-level Keras API.

Here it is the step-by-step explanation of the implementation:

1) *Batch Processing*: During the training, the model is taking online batches of size 128 from over 5,000 subjects. In this case, the composition of the triplet is done by the function in the loss step. Here, the batch generator outputs pairs of Anchor-Positive images.

2) *Embedding Vectors*: The model outputs an embedding vector for each image in the batch. The model used was simpler than the previous case, using a Convolutional Neural Network with a DenseNet121 as a backbone.

3) *Loss Calculation*: For every Anchor-Positive pair in the batch:

- Find a Negative example that is a semi-hard negative. This is an example that has a higher distance from the Anchor than the Positive, but still within a range that poses a challenge for the model to learn from without being too restrictive. If no such negative exists, uses the largest negative distance instead.
- Compute the triplet loss for each valid triplet found (Anchor, Positive, and semi-hard Negative) according to the loss function in Figure 5.

$$d(A, P) < d(A, N) < d(A, P) + \alpha$$

Fig. 5. Semi-hard triplet loss

In other words, the distance between the Anchor and the Negative is greater than the distance between the Anchor and the Positive but still within the margin α .

- It sums these losses together and finally returns the average loss over all the triplets.

4) *Backpropagation and Parameter Updates*: Based on the computed loss, the network weights are updated through backpropagation to bring the embeddings of the Anchor and Positive closer and push the Anchor and semi-hard Negative further apart.

C. ArcFace

ArcFace is a method used in face recognition tasks, the core innovation lies in its loss function, which is designed to enforce a geometric constraint on the angular distance between feature vectors. This helps to improve the discriminative power of the learned features by modifying the standard softmax loss function. The implementation is as follows:

1) *Batch Processing*: For this method, the model is taking online batches of size 64 from over 3,000 subjects. In this case, the output of the batch is the image itself and the one-hot-encoder of the subject id.

2) *Embedding Vectors*: Face images are fed into the network and the network extracts feature vectors from the images.

3) *Loss Calculation*: The angles between the feature vectors and the weight vectors of the corresponding classes are calculated as follows:

- Additive Angular Margin Loss. ArcFace introduces an additive angular margin to the standard softmax loss.

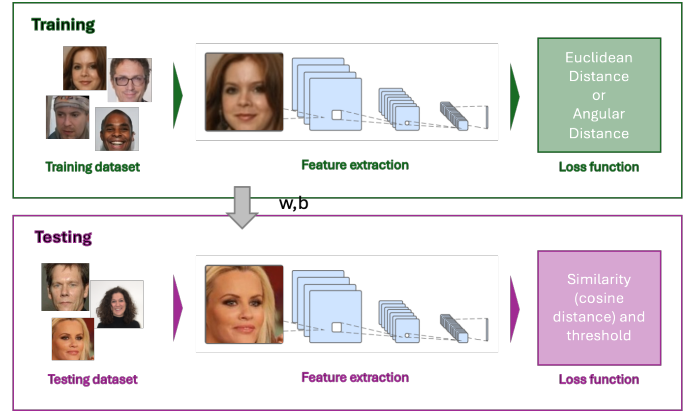


Fig. 7. Face Recognition Pipeline

This margin increases the decision boundary between different classes (identities), making the learned features more discriminative. The formula for the ArcFace loss is as shown in Figure 6.

$$L_3 = -\frac{1}{N} \sum_{i=1}^N \log \frac{e^{s(\cos(\theta_{y_i} + m))}}{e^{s(\cos(\theta_{y_i} + m))} + \sum_{j=1, j \neq y_i}^n e^{s \cos \theta_j}}$$

Here:

- N is the batch size.
- C is the number of classes (identities).
- θ_{y_i} is the angle between the feature vector of the i -th sample and the weight vector of its corresponding class y_i .
- θ_j is the angle between the feature vector and the weight vector of class j .
- s is a scale factor to control the magnitude of the logits.
- m is the additive angular margin.

Fig. 6. ArcFace Formula

- The cosine of the angle between the feature vector and the class weight vector is used, which naturally fits the idea that similar faces should have a smaller angle between their feature vectors.
- Margin Enforcement: By adding the margin m to the angle θ_{y_i} , ArcFace ensures that intra-class variations are reduced (i.e., faces of the same person are closer together in the feature space) while inter-class variations are increased (i.e., faces of different people are further apart).

4) *Backpropagation and Parameter Updates*: The loss is backpropagated through the network to update the weights.

IV. DEVELOPMENT

The three different methods presented earlier were trained and tested according to the face recognition pipeline shown in Figure 7.

No data augmentation was applied to the images, as they are generated for a CNN and available in the DCFace [9] dataset.

It was considered that the noise present was sufficient. For the Triplet Loss, the number of epochs was 2,500, while for the semi-hard variant, it was 5,000. Other tests with different parameters were conducted, but these were the chosen ones. For ArcFace, the number of epochs was much lower at 50 due to the high computational cost. This method was trained on the Kaggle platform using an NVIDIA Tesla P100 accelerator.

The test dataset used in the evaluation comprised 1,000 subjects selected from the CASIA-WebFace [8] dataset. The subjects were divided into two groups:

- **Positive-pairs:** Each positive pair consisted of two distinct images of the same subject, ensuring that the model's ability to correctly identify or verify the same person in different images was tested. These pairs were created by selecting a total of 500 subjects.
- **Negative-pairs:** Each negative pair comprised images of two different individuals. These pairs were formed by randomly selecting images from different subjects within the dataset, to get a total of 500 pairs. The purpose of these pairs was to evaluate the model's effectiveness in distinguishing between different people.

By constructing the test dataset in this manner, the evaluation process was able to rigorously assess both the accuracy of the model in recognizing identical faces and its robustness in discriminating between non-identical faces. This setup ensured a comprehensive evaluation of the face recognition system's performance under realistic conditions.

For the evaluation part, the accuracy, precision and recall was calculated for a range of similarity thresholds. Additionally, for assessing the performance of biometric models, the following error probabilities were used:

- **False Acceptance Rate (FAR):** the percentage of recognitions where someone is wrongly recognised (false acceptance).
- **False Rejection Rate (FRR):** the percentage of recognitions where someone is wrongly not recognised (false rejection).

As the number of false acceptances (FAR) decreases, the number of false rejections (FRR) will increase and vice versa, see Figure 8. The point at which the lines cross also has the name the Equal Error Rate (EER). That is where the percentage of false acceptance and false rejection is equal.

When aiming for the lowest possible FAR, the FRR is likely to become much larger. This means, the more secure the access control, the lower the ease of use because users are mistakenly not recognised by the system.

V. RESULTS AND COMPARISONS

As it was explained before, the performance was evaluated across different decision thresholds. The results are presented in the form of accuracy, precision, recall, False Acceptance Rate (FAR), and False Rejection Rate (FRR) metrics. Below, we analyze these metrics to understand the models' behavior.

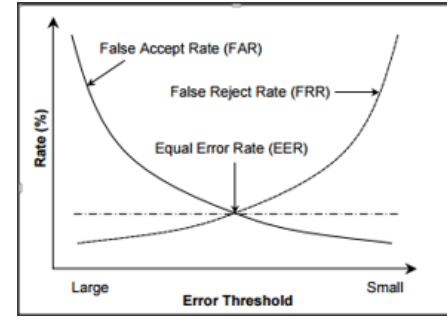


Fig. 8. Relation between FAR and FRR

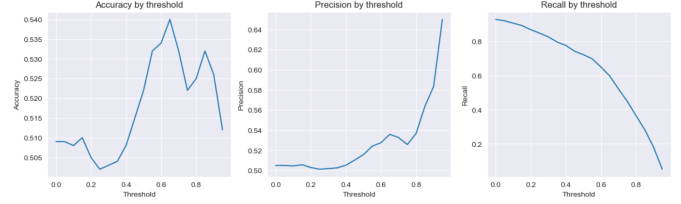


Fig. 9. Accuracy, Precision and Recall curves by similarity threshold for the Triplet Loss method

A. Triplet Loss

Looking at Figures 9 and 11, the accuracy shows that the optimal threshold for the highest overall accuracy is around 0.534, recorded at 0.6. Beyond this point, accuracy slightly decreases again, with a final value of 0.526 at a threshold of 0.9. For precision, higher thresholds reduce the number of false positives, hence improving precision. The model is more confident in its positive identifications as the threshold increases. Precision starts at around 0.505 at a threshold of 0.0 and generally increases with the threshold, reaching 0.583 at a threshold of 0.9. Recall decreases as the threshold increases, lower thresholds result in higher recall, as the model identifies more true positives. However, this also increases the false positives, reducing precision. A high recall at lower thresholds, 0.928 at a threshold of 0.0, indicates the model's sensitivity in detecting true matches.

FAR remains relatively stable around 0.495 at lower thresholds and starts to decrease more significantly beyond the 0.5 threshold, reaching a low of 0.417 at a threshold of 0.9. FRR, on the other hand, generally increases with the threshold, starting from 0.444 at a threshold of 0.0 and rising to 0.485 at a threshold of 0.9.

In general, we can say that the results of this method are not very good and the model probably needs more number of epochs and more triplet combinations to improve.

B. Semi-hard Triplet Loss

Looking at Figures 12 and 14, the accuracy gets a peak of 0.700 at a threshold of 0.2. It decreases steadily beyond this threshold maintaining relatively higher accuracy across thresholds. Precision increases steadily with threshold, and the highest precision of 0.850 at a threshold of 0.9. There are

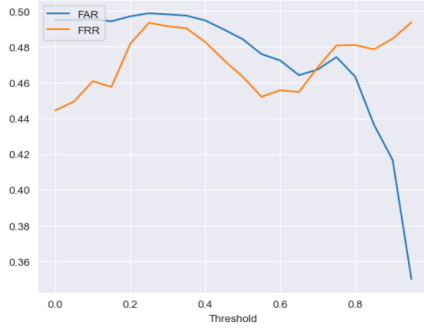


Fig. 10. FAR and FRR curves by similarity threshold for the Triplet Loss method

threshold	Accuracy	Precision	Recall	FAR	FRR
0	0.0	0.509	0.504897	0.928	0.495103
1	0.1	0.508	0.504454	0.906	0.495546
2	0.2	0.505	0.502897	0.868	0.497103
3	0.3	0.503	0.501823	0.826	0.498177
4	0.4	0.508	0.505208	0.776	0.494792
5	0.5	0.522	0.515714	0.722	0.484286
6	0.6	0.534	0.527597	0.650	0.472403
7	0.7	0.532	0.532653	0.522	0.467347
8	0.8	0.525	0.536657	0.366	0.463343
9	0.9	0.526	0.583333	0.182	0.416667

Fig. 11. Value's metrics for the Triplet Loss method

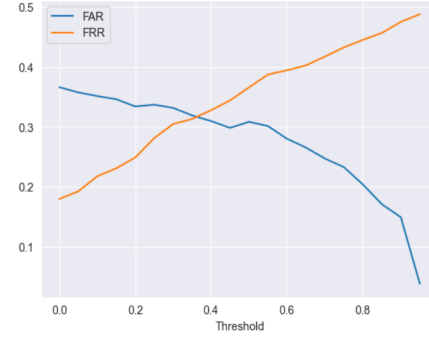


Fig. 13. FAR and FRR curves by similarity threshold for the Semi-hard Triplet Loss method

threshold	Accuracy	Precision	Recall	FAR	FRR
0	0.0	0.689	0.634043	0.894	0.365957
1	0.1	0.695	0.648855	0.850	0.351145
2	0.2	0.700	0.666113	0.802	0.333887
3	0.3	0.681	0.668529	0.718	0.331471
4	0.4	0.681	0.690526	0.656	0.309474
5	0.5	0.658	0.691748	0.570	0.308252
6	0.6	0.643	0.720000	0.468	0.280000
7	0.7	0.625	0.753036	0.372	0.246964
8	0.8	0.593	0.796178	0.250	0.203822
9	0.9	0.547	0.850746	0.114	0.149254

Fig. 14. Value's metrics for the Semi-hard Triplet Loss method

high recall at lower thresholds, starting from 0.894 at 0.0, decreasing as the threshold increases, ending at 0.114 at 0.9.

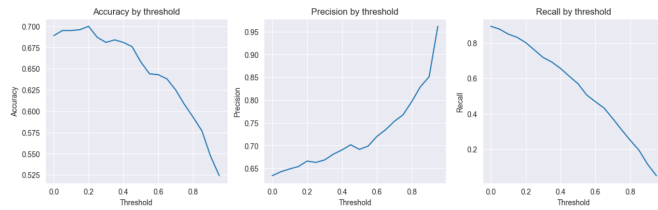


Fig. 12. Accuracy, Precision and Recall curves by similarity threshold for the Semi-hard Triplet Loss method

The FAR decreases as the threshold increases, while the FRR increases. The crossover point or EER is found at a threshold near 0.4.

In this case, the optimal threshold for achieving the highest accuracy is around 0.2, where the model balances accuracy and recall reasonably well. However, for safety reasons, a higher threshold may be preferred, despite the trade-off between recall and accuracy. For example, at a threshold of 0.4, where precision is higher, we would have 127 cases, see Figure 15, where unauthorised people have logged into an account.

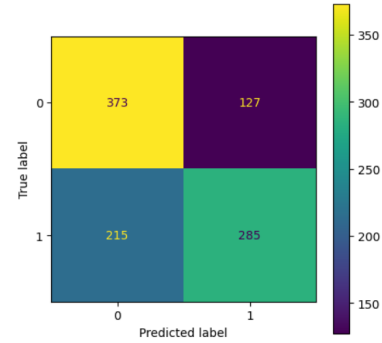


Fig. 15. Confusion matrix with a threshold of 0.5 in the Semi-hard triplet loss model

In comparison, the Semi-hard Triplet Loss model generally provides better performance across most metrics compared to the Triplet Loss method. However, its performance is still insufficient, making it unreliable for use as a dependable facial recognition system.

C. ArcFace

For the ArcFace method, we initially set the loss function with the same feature scale and angular margin as mentioned



Fig. 16. True Positive examples



Fig. 17. False Negative examples

in its paper [6], specifically setting s to 64 and m to 0.5 and using a regularizer L2. However, the results were highly unsatisfactory. The vector embeddings for all images were almost identical, indicating that the method was failing to create distinct spaces for different identities. This failure suggested that the network wasn't learning to differentiate between the features of different faces.

The authors of the ArcFace loss explain that small variations in the feature scale and angular margin can significantly affect the learning process and the performance of the model. These parameters are crucial as they directly influence the stability and convergence during training. Even slight deviations can lead to instability and poor convergence, resulting in suboptimal performance.

In an attempt to resolve these issues, we conducted some experiments with different values for s and m . We hoped that by fine-tuning these parameters we could achieve better separation of the embeddings and improve the overall performance of the model. Despite our tests and adjustments, none of the results were satisfactory. The model continued to produce embeddings that were too similar and failed to effectively discriminate between different faces. This indicated that finding the optimal values for the feature scale and angular range was critical and more complex than originally expected.

Finally, as a test of a real scenario, and as mentioned at the beginning of this report, different images of one of the authors were used to test the best of the three models, the Semi-Hard Triplet Loss, although the results weren't good enough to use it in a real face recognition system for the company. Using 8 image pairs and a threshold of 0.7, Figure 16 shows some of the positive cases detected by the model, while Figure 17 shows some of the cases where the model failed.

VI. CONCLUSION

Although the results are not sufficient to implement any of the methods as a facial login system in production, they

lay the groundwork for a larger project aimed to develop an effective and reliable login system. The experiments and findings, despite not meeting the required performance standards, provided valuable insights and a solid foundation to build upon. They highlighted the potential and feasibility of the approach, indicating that with further research and refinement, a successful facial recognition login system could be achieved. This initial phase demonstrated that continuing the project is not only possible but also promising for future developments.

Looking ahead, there are several planned strategies to present to the company in order to explore ways to improve the effectiveness of the facial recognition model:

- Experimenting with different networks for the backbone model, which could enhance feature extraction and overall performance.
- Using pre-trained weights from datasets more focused on facial images, rather than the generic *imagenet* weights, which would improve the model's ability to distinguish subtle differences between faces.
- Incorporating advanced data augmentation techniques, such as different perspectives of the person and facial detection for cropping and centering the face, can also improve the robustness of the model by providing it with a more diverse set of training examples.
- Additionally, training a base model with real images rather than synthetic ones could significantly improve the authenticity and accuracy of the system.

These future efforts will build on the initial groundwork, and if a model with promising results is found, a larger test using real users' images will be conducted in accordance with the General Data Protection Regulation. This will help evaluate the system's efficiency and ensure a proper balance between secure access and user convenience.

REFERENCES

- [1] CVPR 2024 Challenge: <https://frcsyn.github.io/CVPR2024.html>.
- [2] XING: <https://www.new-work.se/en/career/our-brands/xingbrand>
- [3] Schroff, D. Kalenichenko, and J. Philbin. FaceNet: A Unified Embedding for Face Recognition and Clustering. In CVPR, 2015.
- [4] Alexander Hermans, Lucas Beyer and Bastian Leibe. In Defense of the Triplet Loss for Person Re-Identification. 2017.
- [5] Triplet Loss and Online Triplet Mining: <https://omoindrot.github.io/triplet-loss>
- [6] Deng, Jiankang and Guo, Jia and Yang, Jing and Xue, Niannan and Kotsia, Irene and Zafeiriou, Stefanos. ArcFace: Additive Angular Margin Loss for Deep Face Recognition. IEEE Transactions on Pattern Analysis and Machine Intelligence, 2022, pages: 5962–5979.
- [7] Tensorflow tripletsemihardloss: <https://stackoverflow.com/questions/65579247/how-does-the-tensorflows-tripletsemihardloss-and-triplethardloss-and-how-to-use>
- [8] CASIA-WebFace dataset: <https://github.com/davidsandberg/facenet>
- [9] DCFace dataset: <https://github.com/mk-minchul/dface>